

Introductory Econometrics with Recitation — Quiz 6

ECON2023

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Name: _____ ID: _____

1. (2 points) What is the output of the following code?

```
1 grades <- data.frame(  
2   student = c("A", "B", "C", "D"),  
3   quiz1 = c(80, 75, 90, 65),  
4   quiz2 = c(90, 85, 100, 70)  
5 )  
6 grades_long <- grades %>%  
7   pivot_longer(  
8     cols = c(quiz1, quiz2),  
9     names_to = "quiz",  
10    values_to = "score"  
11  )  
12 dim(grades_long)
```

- A. 8 2
- B. 12 2
- C. 8 3
- D. 12 3

2. (2 points) What is the output of the following code?

```
1 words <- c("cat", "catalog", "mycat")  
2 grepl("^cat$", words)
```

- A. FALSE FALSE FALSE
- B. TRUE FALSE FALSE
- C. TRUE TRUE FALSE
- D. TRUE TRUE TRUE

3. (2 points) What is the output of the following code?

```
1 gsub("C+", "", "ABCDE")
```

- A. ""
- B. "AB"
- C. "ABE"

D. "ABDE"

4. (2 points) Suppose Alice wants to calculate the pseudo R-squared of a probit model. Which function can she use?

- A. `pR2()`
 B. `probit_R2()`
 C. `psuedo_R2()`
 D. `pseudo_Rsquared()`

Please read the following context and answer Questions 5–7.

Context: Hiraku wants to know how much the predicted probability of becoming a high-yield field increases if the field is cultivated with the Almighty Farming Tool. He has a dataset that documents current field information, which contains:

- `crop`: The crop planted in the field.
- `field_size_m2`: The size of the field, measured in square meters.
- `almighty_farming_tool`: = "yes" if the field was cultivated using the Almighty Farming Tool, = "no" otherwise.
- `high_yield_field`: = 1 if the field is a high-yield field, = 0 otherwise.

Hiraku runs regressions where the response variable is `high_yield_field`, and the predictors are `field_size_m2`, `almighty_farming_tool`, and `crop`, using LPM, logit, and probit models. The regression results are shown in the table below by `screenreg()`:

	Model 1: LPM (Robust)	Model 2: Logit	Model 3: Probit
(Intercept)	-0.3916 *** (0.1054)	-13.7660 ** (4.4375)	-7.8376 ** (2.5049)
<code>field_size_m2</code>	0.0001 *** (0.0000)	0.0022 ** (0.0008)	0.0012 ** (0.0005)
<code>almighty_farming_tool</code> yes	0.7891 *** (0.0542)	9.6782 *** (2.8526)	5.4807 *** (1.5591)
<code>crop</code> corn	0.2259 ** (0.0840)	2.5573 (1.6787)	1.5638 (0.9346)
<code>crop</code> potato	0.0473 (0.0793)	0.8728 (1.4063)	0.5826 (0.8023)
<code>crop</code> tomato	0.1784 * (0.0889)	1.5289 (1.4501)	0.9754 (0.8022)
R ²	0.6950		
Adj. R ²	0.6768		
Num. obs.	90	90	90
RMSE	0.2823		
AIC		46.5804	46.0882
BIC		61.5793	61.0871
Log Likelihood		-17.2902	-17.0441
Deviance		34.5804	34.0882
*** p < 0.001; ** p < 0.01; * p < 0.05			

5. (2 points) Which of the following statements is TRUE?
- A. **There are four categories in crop.**
 - B. The table is printed with the argument `include.ci = TRUE`.
 - C. The LPM, logit, and probit models can all be estimated using `lm_robust()`.
 - D. Since `high_yield_field` is a binary variable, the LPM, logit, and probit models all come from the argument `family = binomial(...)`.
6. (2 points) Using the LPM model, what is the predicted probability that a 1000-square-meter potato field cultivated without using the Almighty Farming Tool is a high-yield field?
- A. 0.1473
 - B. 0.0473
 - C. -0.1443
 - D. **-0.2443**
7. (2 points) Assume that the regression results of Model 3 are saved in the variable `probit_high_yield`. Which interval does the output from the following code lie in?
- ```

1 new_field <- data.frame(
2 field_size_m2 = 2500,
3 almighty_farming_tool = "yes",
4 crop = "potato"
5)
6 predict(probit_high_yield, newdata = new_field, type = "response")

```
- A. [0.00, 0.50)
  - B. **[0.50, 0.95)**
  - C. [0.95, 0.975)
  - D. [0.975, 1.00]

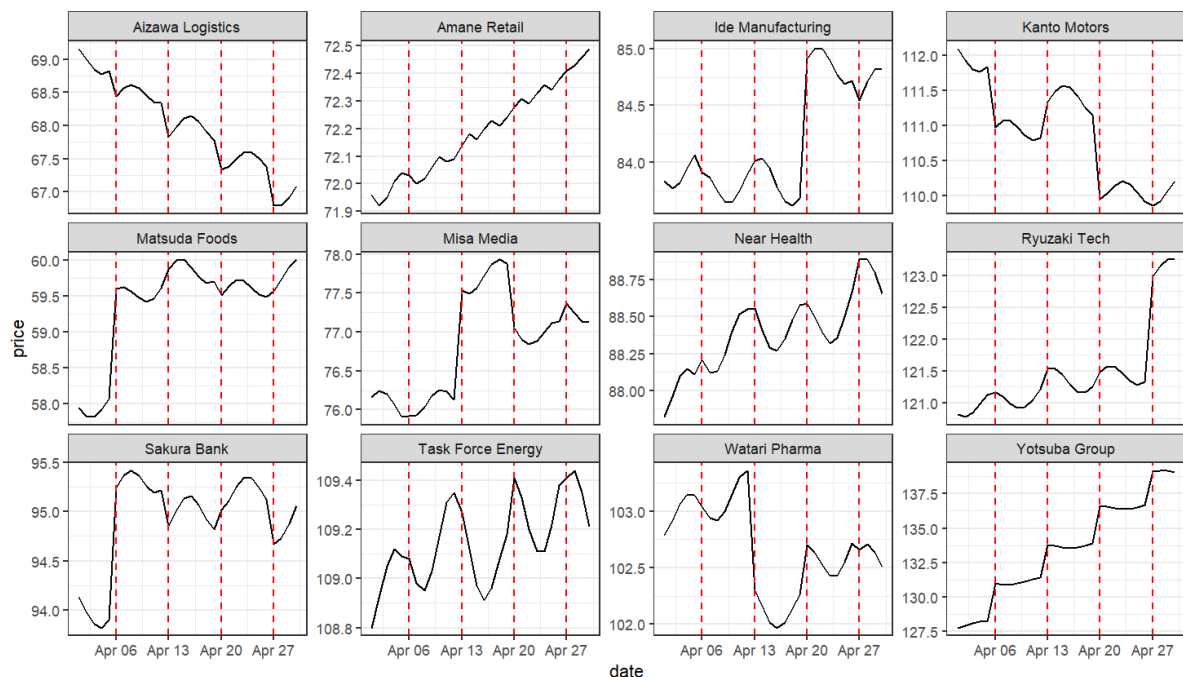
**Please read the content about Week 11's practice on Pages 7-8 to answer Questions 8-10:**

8. (2 points) In the function `convert_time_to_sec()`, the code extracts the second part from `final_time` in `m:ss.s` format. Which of the following pieces of code fails to do the same job?
- A. `sub("^([0-9]):", "", x)`
  - B. `sub(".: ", "", x)`
  - C. `sub(".+: ", "", x)`
  - D. **`sub("^+", "", x)`**
9. (2 points) Which of the following statements about `geom_smooth()` is TRUE?

- A. The `formula =` argument fits a constant line  $y \sim 1$  by default.
- B. The argument `se = FALSE` hides the standard error text calculated from the formula argument in the plot.
- C. The number of smooth lines shown in the graph depends on the argument `color = runner`.
- D. To make `geom_smooth()` work, we need to add `geom_point()` before it.
10. (2 points) In Lines 36–41, we run a second-order polynomial regression. Which of the following statements about the regression is FALSE?
- A. It is reasonable to consider a polynomial regression here since the increase in running time by distance is likely not linear.
- B. The `time_distance_plot` created in Line 25 plots a polynomial relationship. We expect the curves in the plot to be convex.
- C. The regression may suffer from omitted variable bias because it only uses the training data to predict the outcome. Therefore, the outcomes predicted from the model will be downward biased.
- D. Given that Mejiro McQueen is better at long-distance races, the estimated coefficient on `distance_km2` for her will likely be smaller than Tokai Teio's.

Please read the content about Week 13's practice on Pages 8-10 to answer Questions 11-15:

11. (2 points) The figure below shows the plot created in Lines 11-18. Which of the following statements about this graph is FALSE?



- A. Yotsuba Group looks the most suspicious because there are repeated stock price surges after each death date.
  - B. From the graph, we can infer that the dataset documents stock prices for each company over the same date range, so replacing "free\_y" with "free" will not affect the output graph.
  - C. **According to the graph, to visually highlight the increase in stock price, it would be better to set scales = "fixed" so the y-axes are shown in the same numeric range.**
  - D. There are exactly 4 vertical lines in each subgraph because the length of death\_dates is 4.
12. (2 points) Which of the following statements about mutate() used in Lines 20-26 is TRUE?
- A. **If a row has value 1 in column after\_arima\_death, then its value in column after\_kamei\_death must also be 1.**
  - B. Moving %>% from the end of Line 20 to the beginning of Line 21 also works.
  - C. Replacing = with <- inside mutate() also works.
  - D. Replacing Line 22 with the following code also works:
- ```

1 if(date >= as.Date("2026-04-06")){after_kamei_death <- 1}
2 else{after_kamei_death <- 0}

```
13. (2 points) Inside the for loop, Light estimates a TWFE regression for each company. Which of the following statements about the regression is TRUE?
- A. In each iteration of the loop, the regression drops all observations where target_company == 0, estimating the coefficients using only data from the target company.
 - B. The inclusion of both company and date fixed effects completely eliminates the threat of omitted variable bias.
 - C. The estimated coefficients capture the one-day price jump on the exact date of each death.
 - D. **The baseline trend compared against each target company is actually different in each loop because the control group composition shifts.**
14. (2 points) Which of the following statements about Lines 32-65 is TRUE?
- A. Using suspect_results\$company[1] in Line 65 is unreliable because the selected value may depend on the order of the rows in the raw dataset.
 - B. **Removing the pivot_wider() step in Lines 59-62 does not affect the selection of most_suspicious_company or the analysis below.**
 - C. The code also works if we replace (c in companies) with (i in 1:12) in Line 32.

- D. We can replace `rbind()` with `cbind()` in Line 51.
15. (2 points) Which of the following statements about Lines 70-80 is FALSE?
- A. Using `vcov = "iid"` calculates standard errors under the assumption of homoscedasticity.
 - B. Although `summary(reg_suspicious, vcov = ~ company)` calculates standard errors clustered by `company`, it is not guaranteed that these standard errors will be weakly larger than those calculated by `summary(reg_suspicious, vcov = "iid")`.
 - C. Since the same regression has been run in the loop above, we can also replicate the summary outputs by extracting stored results from `reg_twfe` and `suspect_results`.
 - D. It is not recommended to replace `:` with `*` in Lines 71-74; it will make `fixest` return a collinearity message in this regression.

Below is the context and the answer code for **Week 11's practice**. Please read the content and answer **Questions 8-10**:

Context: Tokai Teio and Mejiro McQueen are preparing for the Emperor's Prize, one of the highest-level races in horse racing.

Tokai Teio has been outstanding at middle distances and remains undefeated, but she suffered a fracture a few months ago and is now preparing for her comeback. Mejiro McQueen is known for her strength at longer distances and is seeking to defend her title in the Emperor's Prize. Their showdown in the Emperor's Prize has become the center of attention for horse racing fans.

In this practice, we use the training data of Tokai Teio and Mejiro McQueen to predict the race outcome. The dataset contains:

- `date`: The date of the training session.
- `runner`: The runner's name (Tokai Teio or Mejiro McQueen).
- `distance_m`: The distance in meters.
- `final_time`: The recorded finishing time, stored as a character string in `m:ss.s` format.

Given that the Emperor's Prize is a 3200-meter race, who is predicted to win the Emperor's Prize based on the training data?

Answer code:

```

1 library(tidyverse)
2 library(estimatr)
3
4 pretty_derby_raw <- read.csv("data/pretty_derby.csv")
5 pretty_derby_tidy <- pretty_derby_raw
6
7 convert_time_to_sec <- function(x) {
8   min_part <- as.numeric(sub(":.*", "", x))
9   sec_part <- as.numeric(sub(".*:", "", x))
10  min_part * 60 + sec_part
11 }
12
13 fracture_date <- as.Date("1992-01-10")
14
15 pretty_derby_tidy <- pretty_derby_tidy %>%
16   mutate(
17     date = as.Date(date),
18     race_time_sec = convert_time_to_sec(final_time),
19     distance_km = distance_m / 1000,
20     after_fracture = if_else(
21       runner == "Tokai Teio" & date > fracture_date, 1, 0
22     )
23   )

```

```

24
25 time_distance_plot <- pretty_derby_tidy %>%
26   ggplot(aes(x = distance_m, y = race_time_sec, color = runner)) +
27   geom_point() +
28   geom_smooth(
29     method = "lm",
30     formula = y ~ x + I(x^2),
31     se = FALSE
32   )
33
34 print(time_distance_plot)
35
36 model_pretty_derby <- lm_robust(
37   race_time_sec ~ runner * distance_km + runner * I(distance_km^2) +
38   ~ after_fracture,
39   data = pretty_derby_tidy
40 )
41 summary(model_pretty_derby)

```

Below is the context and the answer code for **Week 13's practice**. Please read the content and answer **Questions 11-15**:

Context: Light and L are tracking a new Killer, a person who owns a Death Note and can cause people to die of cardiac arrest by writing their names in it. After this new Killer appears, Light notices something unusual. The Killer does not write only the names of criminals, but also writes the names of top managers from large companies. Light begins to suspect that the new Killer may work for a large company and may be using the Death Note for private profit. If this hypothesis is correct, then the stock price of the company the Killer works for should increase after each suspicious death.

To investigate this possibility, Light collects daily stock prices for several companies and tracks how their prices change after each death. The dataset contains:

- **company:** The company's name.
- **date:** The date.
- **price:** The stock price.

The suspicious deaths occurred once per week:

Event	Date	Top manager killed
1	2026-04-06	Ryo Kamei
2	2026-04-13	Naomi Ogawa
3	2026-04-20	Keiko Arima
4	2026-04-27	Shun Takeda

If Light's hypothesis is correct, then there should be repeated stock price surges in the Killer's company. Follow the instructions below to help Light test this hypothesis.

Answer code:

```

1 library(tidyverse)
2 library(fixest)
3
4 death_note_raw <- read.csv("data/death_note.csv")
5
6 death_note_tidy <- death_note_raw %>%
7   mutate(date = as.Date(date))
8
9 death_dates <- as.Date(c("2026-04-06", "2026-04-13", "2026-04-20", "2026-04-27"))
10
11 ggplot(death_note_tidy, aes(x = date, y = price)) +
12   geom_line() +
13   geom_vline(
14     xintercept = death_dates,
15     color = "red",
16     linetype = "dashed"
17   ) +
18   facet_wrap(~ company, scales = "free_y")
19
20 death_note_tidy <- death_note_tidy %>%
21   mutate(
22     after_kamei_death = ifelse(date >= as.Date("2026-04-06"), 1, 0),
23     after_ogawa_death = ifelse(date >= as.Date("2026-04-13"), 1, 0),
24     after_arima_death = ifelse(date >= as.Date("2026-04-20"), 1, 0),
25     after_takeda_death = ifelse(date >= as.Date("2026-04-27"), 1, 0)
26   )
27
28 companies <- unique(death_note_tidy$company)
29
30 suspect_results <- data.frame()
31
32 for (c in companies) {
33   temp <- death_note_tidy %>%
34     mutate(target_company = ifelse(company == c, 1, 0))
35
36   reg_twfe <- feols(
37     price ~ target_company:after_kamei_death +
38       target_company:after_ogawa_death +
39       target_company:after_arima_death +
40       target_company:after_takeda_death |
41     company + date,
42     data = temp
43   )
44

```

```

45 temp_result <- data.frame(
46   company = c,
47   event = death_dates,
48   estimate = as.numeric(reg_twfe$coefficients)
49 )
50
51 suspect_results <- rbind(suspect_results, temp_result)
52 }
53
54 suspect_results <- suspect_results %>%
55   group_by(company) %>%
56   mutate(
57     mean_increase = mean(estimate)
58   ) %>%
59   pivot_wider(
60     names_from = event,
61     values_from = estimate
62   ) %>%
63   arrange(desc(mean_increase))
64
65 most_suspicious_company <- suspect_results$company[1]
66
67 death_note_suspicious <- death_note_tidy %>%
68   mutate(target_company = ifelse(company == most_suspicious_company, 1, 0))
69
70 reg_suspicious <- feols(
71   price ~ target_company:after_kamei_death +
72     target_company:after_ogawa_death +
73     target_company:after_arima_death +
74     target_company:after_takeda_death |
75     company + date,
76   data = death_note_suspicious
77 )
78
79 summary(reg_suspicious, vcov = "iid")
80 summary(reg_suspicious, vcov = ~ company)

```

Answer Key

1.C 2.B 3.D 4.A 5.A 6.D 7.B 8.D 9.C 10.C 11.C 12.A 13.D 14.B 15.C